

Creative Coding & Computational Design

Interactiv · Curriculum Overview

Programme purpose

A design-technology programme that helps students use computation, interaction and generative systems as creative tools. Students learn computational thinking through making, moving from ideas and algorithms to systems, interaction and experiences.

Learning progression

Idea → Algorithm → System → Interaction → Experience

The programme can begin with visual and block-based approaches before introducing professional creative coding and interactive technology tools.

15 Core Concepts

01 · Sequence & Logic	Break a creative idea into ordered steps.
02 · Patterns & Repetition	Use loops and repetition to create visual systems.
03 · Variables & Parameters	Make designs dynamic and controllable.
04 · Conditions & Rules	Create behaviour based on decisions and logic.
05 · Randomness & Variation	Generate unexpected visual outcomes.
06 · Coordinates & Transformation	Position, scale, rotate and move.
07 · Functions & Modular Thinking	Build reusable creative systems.
08 · Events & Interaction	Make experiences respond to user input.
09 · State & Behaviour	Design systems that change over time.
10 · Data as a Creative Material	Turn information into visual and interactive forms.
11 · Sound & Multimodal Interaction	Connect audio with visual behaviour.

12 · Physical Computing	Connect sensors, electronics and code.
13 · Computer Vision & Tracking	Respond to people and movement.
14 · Spatial & Immersive Computing	Design interaction across physical space.
15 · Systems Thinking & Experience Design	Combine technology into a coherent experience.

Programme formats

2-Day Workshop · Introduction to visual systems, computational thinking and a first interactive/generative piece.

6-Week Module · Structured introduction to creative coding, generative systems, interaction and a portfolio project.

14-Week Semester · Full studio programme covering foundations, systems, interaction, extensions and a capstone.

Programme fit

Especially relevant to Animation & Motion Design, Interaction Design & UI/UX, Communication & Visual Design, and Digital Media & Game Design.

This document is an overview. Detailed session plans, project briefs, assessment and tool choices can be adapted to the institution's academic framework.